

- Q.25 Describe the constructional features of an automatic moulding machine. (CO2)
- Q.26 Write the difference between pneumatic and hydraulic moulding machines. (CO2)
- Q.27 Explain the importance of cooling system in injection moulding. (CO3)
- Q.28 Write the application of injection moulding. (CO3)
- Q.29 Write the advantages of injection moulding. (Co3)
- Q.30 Explain the factors affecting the quality of compression moulded parts. (CO4)
- Q.31 Explain the types of transfer moulds. (CO4)
- Q.32 Discuss the advantages and disadvantages of transfer moulding. (CO4)
- Q.33 Write the components of a transfer moulding machine. (CO4)
- Q.34 Write the advantages and limitations of using pre-hardened steel in mould manufacturing. (CO5)
- Q.35 Explain the importance of lubrication in mould maintenance. (CO6)

#### SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain the factors affecting the quality of compression moulded parts. (CO4)
- Q.37 Explain the working principle of screw type injection moulding machine. (CO3)
- Q.38 Discuss common mould defects caused due to poor maintenance and suggest corrective actions. (CO6)

**Note :** Course Outcome (CO) mentioned in the question paper is for official purpose only.

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#### 5th Sem / (Tool & Die) Sub : Plastic Mould Design & Drawing

Time : 3Hrs.

M.M. : 100

#### SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which material is commonly used for making pattens? (CO1)
- a) Rubber b) Plastic  
c) Wood d) Glass
- Q.2 Draft allowance is provided on a pattern to \_\_\_\_\_. (CO1)
- a) Increase weight  
b) Improve appearance  
c) Remove pattern easily from mould  
d) Reduce shrinkage
- Q.3 The jolt-squeeze moulding machine combines \_\_\_\_\_. (CO2)
- a) Rolling and squeezing  
b) Jolting and squeezing  
c) Vibrating and rolling  
d) Slush and squeeze
- Q.4 Which moulding machine is best suited for large moulds?
- a) Bench moulding machine  
b) Jolt machine  
c) Sand slinger  
d) Squeeze machine

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- Q.5 The fixed half of an injection mould is attached to the \_\_\_\_\_ (CO3)  
 a) Moving Platen                      b) Ejector Plate  
 c) Fixed platen                        d) Core plate
- Q.6 The function of a runner in injection moulding is to \_\_\_\_\_ (CO3)  
 a) Cool the plastic  
 b) Guide molten plastic into cavities  
 c) Eject the product  
 d) Clamp the mould
- Q.7 The main advantages of compression moulding is \_\_\_\_\_ (CO4)  
 a) High production rate  
 b) Low mould cost  
 c) No flash formation  
 d) Fully automatic Process
- Q.8 Transfer moulding is suitable for moulding \_\_\_\_\_. (Co4)  
 a) Very large flat parts  
 b) Simple shapes only  
 c) Intricate shapes with inserts  
 d) Pipes
- Q.9 The most commonly used material for injection mould cavities is \_\_\_\_\_. (CO5)  
 a) Mild steel                              b) High carbon steel  
 c) Tool steel                                d) Cast iron
- Q.10 For proto type mould, which material is often used? (CO6)  
 a) Copper                                      b) Aluminum  
 c) Titanium                                    d) Cast

## SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Pattern making is an important step in \_\_\_\_\_. (CO1)  
 Q.12 The extra material added to a casting for machining is called \_\_\_\_\_. (CO1)  
 Q.13 Cooling channels in injection mould are used to \_\_\_\_\_. (CO2)  
 Q.14 The clamping force in injection moulding prevents \_\_\_\_\_. (CO2)  
 Q.15 \_\_\_\_\_ which material is suitable for high-volume production moulds? (CO3)  
 Q.16 Core pins are generally made from \_\_\_\_\_. (CO4)  
 Q.17 Beryllium copper is used in moulds because of its \_\_\_\_\_. (CO5)  
 Q.18 The mould base is generally made of \_\_\_\_\_. (CO5)  
 Q.19 Rust formation in moulds is mainly caused by \_\_\_\_\_. (CO6)  
 Q.20 \_\_\_\_\_ Flash formation in moulded parts may indicate \_\_\_\_\_. (CO6)

## SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write a short note on casting. (CO1)  
 Q.22 Explain the relationship between casting and moulding process. (CO1)  
 Q.23 Describe the advantages and limitations of sand moulding process. (CO2)  
 Q.24 Explain the importance of moulding machines in mass production. (CO2)